

skrub Example Opportunities from Time-Series & Explainability Galleries — Conclusions

Synthesis of the five candidate analyses (2026-09-22): - `aeon-to-skrub-rewrite-candidates.md`
- `tslearn-to-skrub-rewrite-candidates.md` - `skforecast-to-skrub-rewrite-candidates.md`
- `mapie-to-skrub-rewrite-candidates.md` - `shapash-to-skrub-rewrite-candidates.md`

Libraries fall into three relationship classes

1. **Forecasting engines skrub complements** — `skforecast` (same tabular paradigm), plus the forecasting slices of `aeon` and `tslearn`. The libraries' models stay as the prediction nodes; `skrub`'s role is building the tables they assume (aligned exog, calendar features, entity attributes, encoded histories) from messy multi-table sources, plus time-aware splits and structural search around them. A pure `skrub` tabular pipeline appears only as a rival branch inside `choose_from` — searched alongside the native model, never the headline replacement.
2. **Algorithm-internals libraries** — `aeon`, `tslearn`. `skrub` should *not* rewrite their clustering/shapelets/metrics/segmentation material; the honest plays are wrappers (`aeon` rocket / SAX strings / `tsfresh` inside `ApplyToSubFrame`, `choose_from` structure search) and conceptual contrasts.
3. **Downstream layers skrub feeds** — `MAPIE` (conformal prediction), `shapash` (attribution). Integration pattern is `Mapie*(SkrubLearner)` and `shapash.compile(postprocess=...)` over `skrub` encoders; `skrub` contributes the data wrangling and the encoder→raw-column bridge.

Positioning principle for all three classes: `skrub` is the **data & orchestration layer** — join, clean, encode, split, search, report — and the numeric model (`aeon` forecaster, `tslearn` VARIMA, `skforecast` engine, HGB) stays whatever library it came from.

Cross-cutting finding: one script rules them all

The **DataOps timestamped-forecasting pipeline** (lags/rolling features via `AggJoiner/apply_func`, calendar via `DatetimeEncoder`, time-aware splits via `mark_as_X(cv=Splitter())`, `freeze_after_fit`, search via `ParamSearch` — all already proven in `/Users/rcap/work/sessionization/main.py`) is the shared backbone inside *all five* galleries (`aeon` forecasting, `tslearn` VARIMA, `skforecast` intro/energy/global, `MAPIE` ts-tutorial/EnbPI/Zaffran, `shapash` domain05). Build it once, cut five gallery-flavored variants — each keeps the gallery's native model as the head node (rocket baseline, DTW-kNN/VARIMA, `skforecast` engine, `MapieRegressor`, SHAP layer) with the `skrub` tabular path demoted to one `choose_from` branch among them.

Shortlist (ranked across all galleries)

#	Example	Uses	Why it wins
1	Forecasting with DataOps on <code>fetch_electricity_forecasting/fetch_electricity_forecasting</code>	AggJoiner, DatetimeEncoder, splitter, <code>make_randomized_search</code>	Anchors 5 galleries; swappable head estimator — (aeon/skforecast) or tabular chosen by <code>choose_from</code>
2	SAX/shapelets/rocket/twosh → series-as-a-column via <code>ApplyToSubFrame</code>	<code>twosh</code> transformer, tabular_pipeline, <code>choose_from</code>	Unique cross-library story (“your TS featurizer is a column transformer”)
3	What-if / recourse by re- <code>skb.eval</code> on modified graphs	DataOps graph editing	Only skrub can do it; shapash structurally can’t; genuinely novel
4	Structural pipeline search vs Grid-SearchCV/backtesting loops	ParamSearch/Optuna, <code>cross_validate</code> , <code>full_report</code>	Upgrades aeon, <code>knnts_sklearn</code> , MAPIE <code>nested-cv</code> , skforecast tuning in one stroke
5	Multi-table exogenous data (energy demand + weather + holidays)	Joiner/fuzzy_joiner, MultiAggJoiner, TextEncoder/LLMEncoder	skrub’s home turf; skforecast explicitly can’t do dirty exog
6	Explaining original columns through encoder postprocess	TableVectorizer + shapash postprocess, column_associations	Required bridge for any skrub+shapash user; reuses #1
7	Group-conditional CP with groups mined from messy columns	GapEncoder/deduplicate + Mondrian/ResNormalised	Explicit conditioning variables derived from free text
8	Categorical-series forecasting + Titanic end-to-end	ToCategorical, Cleaner, encoders	Cheapest to build; squarely skrub’s strengths

Explicit non-goals

- Rewrites of aeon classification families, tslearn metrics/clustering/matrix-profile/barycenters, MAPIE paper reproductions and exchangeability tests, shapash backends (SHAP/LIME/ShapIQ) and plot mechanics: pure algorithmics or synthetic-data theory demos where skrub adds nothing. Attempting them would misrepresent skrub as a numeric time-series or attribution library.

Suggested production order

1–2 from the shortlist (forecasting script + series-as-column wrapper), then reuse their scaffolding for 4 (search) and 6/7 (the two MAPIE/shapash integrations); 3 is the standalone differentiator worth doing early; 5 and 8 are cheap fillers once 1 exists.